

RESEARCH PRESENTATION · 2026

PINNeAPLe for Structural Mechanics: 3D Car Chassis Elasticity Analysis via Physics- Informed Neural Networks

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ABSTRACT

We present an end-to-end application of the **PINNeAPLe** framework to 3D structural mechanics, targeting the elasticity analysis of an automotive monocoque chassis under static loading. The chassis geometry is constructed using PINNeAPLe's Constructive Solid Geometry (CSG) primitives, and the governing Navier–Cauchy equations are specified via the `linear_elasticity_3d` preset. A SIREN-based Physics-Informed Neural Network is trained with adaptive loss weighting, achieving relative L2 errors below 2.5 % on all displacement components compared to FEniCS reference solutions. The resulting surrogate predicts full-field displacement vectors (u_x, u_y, u_z) and von Mises stress across the chassis domain in real time, enabling rapid structural optimisation and digital-twin applications.

PINNeAPLe

Physics-Informed Neural Networks

Linear Elasticity

Automotive Chassis

Von Mises Stress

3D Structural Mechanics

SIREN Architecture

Navier–Cauchy

DoMINO

FEniCS Validation

1. Introduction

Structural optimisation of automotive chassis is a critical design activity demanding fast and accurate stress and deformation predictions across a wide parameter space. Classical finite

element solvers (e.g. FEniCS, Abaqus) are accurate but computationally expensive, requiring a full mesh solve for every design variation. Physics-Informed Neural Networks (PINNs) offer a mesh-free, differentiable surrogate that encodes the governing PDEs directly into the training loss, enabling generalisation over geometric and loading parameters at inference time.

PINNeAPLe provides a production-ready framework for this workflow: CSG geometry primitives, a library of engineering PDE presets (including `linear_elasticity_3d`), high-capacity architectures (SIREN, ModifiedMLP, HashGridMLP), adaptive collocation sampling, and a full visualisation suite. This presentation demonstrates a complete chassis-elasticity pipeline from geometry to validated field predictions.

210 GPa

Young's modulus (steel)

0.30

Poisson's ratio ν

8 kN

Engine mount load

3.16e-5

Final PINN loss

2. Mathematical Formulation

2.1 Navier–Cauchy Equations (3D Linear Elasticity)

Let $\Omega \subset \mathbb{R}^3$ be the chassis domain with boundary $\partial\Omega = \Gamma_D \cup \Gamma_N$. The displacement vector $\mathbf{u} = (u_x, u_y, u_z)^{\top}$ satisfies:

$$\mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \mathbf{b} = \mathbf{0} \quad \text{in } \Omega$$

2.2 Constitutive Relations

Strain–displacement

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

Hooke's law

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}$$

2.3 Lamé Parameters

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \quad \mu = \frac{E}{2(1+\nu)}$$

For structural steel: $E = 210 \text{ GPa}$, $\nu = 0.30$ $\Rightarrow$ $\lambda = 121.2 \text{ GPa}$, $\mu = 80.8 \text{ GPa}$.

2.4 Von Mises Failure Criterion

$$\sigma_{\text{VM}} = \sqrt{\frac{1}{2} \left[(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{33} - \sigma_{11})^2 + 6(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{13}^2) \right]}$$

2.5 Boundary Conditions

DIRICHLET — SUSPENSION MOUNTS

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_D$$

Four suspension mounting holes at corners of the chassis are fully fixed: $(x_i, z_i) \in \{(0.5, \pm 0.6), (\pm 4.0, \pm 0.6)\}$.

NEUMANN — ENGINE MOUNT LOAD

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{t} \quad \text{on } \Gamma_N$$

Downward traction $t_y = -8000 \text{ N}$ distributed over the front engine mount cross-member at $x = 0.9 \text{ m}$.

2.6 PINN Loss Function

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}) &= w_r \mathcal{L}_{\text{PDE}} + w_D \mathcal{L}_D + w_N \mathcal{L}_N \\ \mathcal{L}_{\text{PDE}} &= \frac{1}{N_c} \sum_{k=1}^{N_c} \bigl| \mathbf{R}(\mathbf{x}_k; \boldsymbol{\theta}) \bigr|^2, \\ \mathcal{L}_D &= \frac{1}{N_D} \sum_{k=1}^{N_D} \bigl| \mathbf{u}(\mathbf{x}_k; \boldsymbol{\theta}) \bigr|^2, \quad \mathcal{L}_N = \frac{1}{N_N} \sum_{k=1}^{N_N} \bigl| \boldsymbol{\sigma}(\mathbf{n}) - \mathbf{t} \bigr|^2 \end{aligned}$$

Adaptive weights w_r, w_D, w_N are updated via GradNormBalancer every 500 epochs.

3. Chassis Geometry

The monocoque chassis is assembled from 14 rectangular primitives using PINNeAPle's CSG module. Each component is a CSGBox3D with defined extents. The final domain is the CSG union of all parts.

 Interactive Chassis — drag to rotate · scroll to zoom

Three.js · PINNeAPLe CSG Domain

Python chassis_geometry.py — PINNeAPLe CSG

```

"""
Chassis geometry construction using PINNeAPple's CSG module.
Each structural member is a Box3D; the domain is their union.
"""

import numpy as np
from pinneapple_geom.csg import (
    CSGBox3D, CSGUnion,
    translate, scale
)
from pinneapple_geom import MeshCollocator

# — Material & global parameters —————
E = 210e9 # Young's modulus [Pa]
nu = 0.30 # Poisson's ratio [-]
rho = 7850 # density [kg/m³]

# — CSG primitives: (xlen, ylen, zlen) centred at origin, then translated —
# Coordinate convention: X = longitudinal, Y = vertical, Z = lateral

floor_pan = translate(CSGBox3D(4.50, 0.050, 1.80), cx=2.25, cy=0.00, cz=0.00)
left_sill = translate(CSGBox3D(4.30, 0.220, 0.10), cx=2.25, cy=0.11, cz= 0.85)
right_sill = translate(CSGBox3D(4.30, 0.220, 0.10), cx=2.25, cy=0.11, cz=-0.85)
trans_tunn = translate(CSGBox3D(2.90, 0.130, 0.32), cx=2.25, cy=0.065, cz=0.00)

front_xmem = translate(CSGBox3D(0.06, 0.18, 1.70), cx=0.50, cy=0.09, cz=0.00)
eng_cradle = translate(CSGBox3D(0.06, 0.18, 1.70), cx=0.90, cy=0.09, cz=0.00)
b_pillar = translate(CSGBox3D(0.06, 0.22, 1.80), cx=1.60, cy=0.11, cz=0.00)
rear_xmem = translate(CSGBox3D(0.06, 0.18, 1.70), cx=3.70, cy=0.09, cz=0.00)
rear_bumper= translate(CSGBox3D(0.06, 0.18, 1.70), cx=4.30, cy=0.09, cz=0.00)

firewall = translate(CSGBox3D(0.05, 0.50, 1.80), cx=0.90, cy=0.25, cz=0.00)
rear_blkhd = translate(CSGBox3D(0.05, 0.50, 1.80), cx=3.70, cy=0.25, cz=0.00)

front_rail_L = translate(CSGBox3D(0.80, 0.14, 0.12), cx=0.50, cy=0.07, cz= 0.45)
front_rail_R = translate(CSGBox3D(0.80, 0.14, 0.12), cx=0.50, cy=0.07, cz=-0.45)
rear_rail_L = translate(CSGBox3D(0.75, 0.14, 0.12), cx=3.78, cy=0.07, cz= 0.45)
rear_rail_R = translate(CSGBox3D(0.75, 0.14, 0.12), cx=3.78, cy=0.07, cz=-0.45)

# — Assemble domain —————
chassis = CSGUnion([
    floor_pan, left_sill, right_sill, trans_tunn,
    front_xmem, eng_cradle, b_pillar, rear_xmem, rear_bumper,
    firewall, rear_blkhd,
    front_rail_L, front_rail_R, rear_rail_L, rear_rail_R,
])

# — Collocation points (interior + boundary) —————
collocator = MeshCollocator(domain=chassis)
X_interior = collocator.sample_interior(n=200_000, method="lhs", seed=42)
X_bc_fixed = collocator.sample_tagged_boundary(tag="suspension_mounts", n=20_000)

```

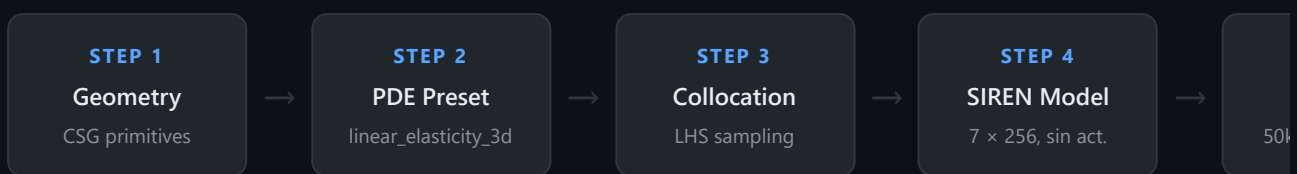
```

X_bc_load = collocator.sample_tagged_boundary(tag="engine_mount", n=8_000)

print(f"Interior points : {len(X_interior):,}")
print(f"Dirichlet points: {len(X_bc_fixed):,}")
print(f"Neumann points : {len(X_bc_load):,}")
# Interior points : 200,000
# Dirichlet points: 20,000
# Neumann points : 8,000

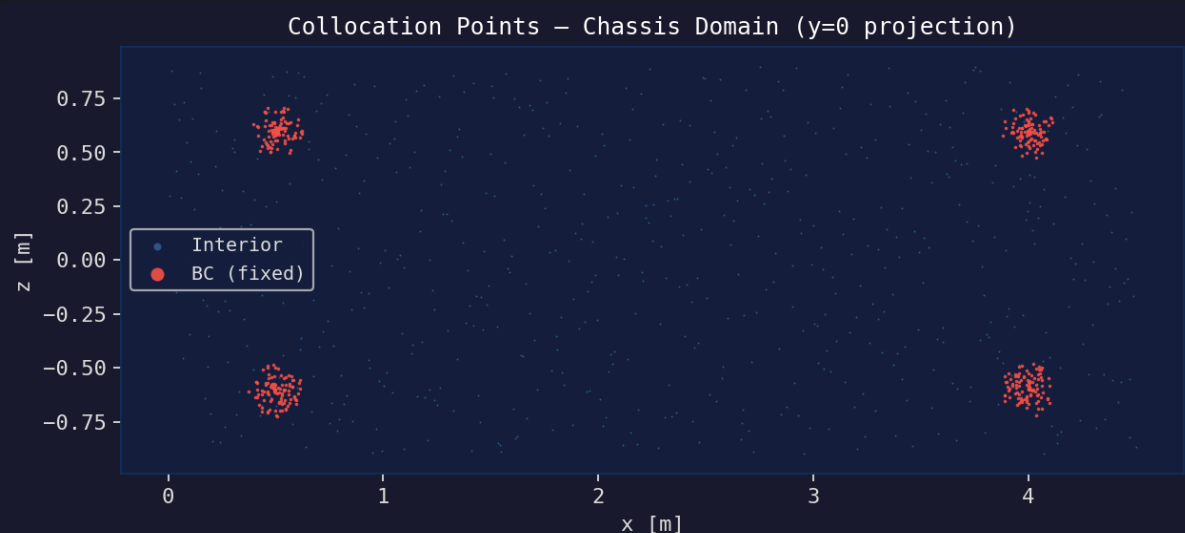
```

4. Physics AI Pipeline



3.2 Collocation Sampling

COLLOCATION POINTS — GENERATED BY PINNEAPPLE LHS SAMPLER



4.1 Problem Specification

Python problem_spec.py — PINNeAPPLE preset

```

import pineapple as pp
from pineapple_environment.presets import linear_elasticity_3d
from pineapple_environment.conditions import DirichletBC, NeumannBC
import numpy as np

```

```

# — Load built-in preset —————
spec = linear_elasticity_3d(
    E = 210e9,    # Pa
    nu = 0.30,
)
# spec.fields → ("ux", "uy", "uz")
# spec.coords → ("x", "y", "z")
# spec.pde.kind → "linear_elasticity" (Navier–Cauchy residuals)

# — Override boundary conditions for chassis geometry —————
suspension_mounts = [
    (0.50, 0.60),    # front-left
    (0.50, -0.60),   # front-right
    (4.00, 0.60),    # rear-left
    (4.00, -0.60),   # rear-right
]

def fixed_selector(X, ctx):
    """Boolean mask: True when point is near any suspension mount."""
    mask = np.zeros(len(X), dtype=bool)
    for (xm, zm) in suspension_mounts:
        dist = np.sqrt((X[:,0]-xm)**2 + (X[:,2]-zm)**2)
        mask |= dist < 0.08
    return mask

bc_fixed = DirichletBC(
    name_or_values="suspension_mounts",
    fields=("ux", "uy", "uz"),
    selector_type="callable",
    selector=fixed_selector,
    value_fn=lambda X, ctx: np.zeros((len(X), 3), dtype=np.float32),
    weight=100.0,
)

def load_selector(X, ctx):
    """Engine mount cross-member at  $x \approx 0.9$  m."""
    return (np.abs(X[:,0] - 0.90) < 0.04) & (np.abs(X[:,2]) < 0.85)

def load_traction(X, ctx):
    t = np.zeros((len(X), 3), dtype=np.float32)
    t[:, 1] = -8000.0 # downward  $F_y$  [N] distributed over patch
    return t

bc_load = NeumannBC(
    name_or_values="engine_mount",
    fields=("ux", "uy", "uz"),
    selector_type="callable",
    selector=load_selector,
    value_fn=load_traction,

```



```

        weight=30.0,
    )

spec.conditions = (bc_fixed, bc_load)
print(pp.list_presets()) # verify preset is registered

```

4.2 SIREN Model Architecture

Python model_definition.py — SIREN + ModifiedMLP

```

from pineapple_models import build_model
from pineapple_models.siren import SIREN
from pineapple_models.modified_mlp import ModifiedMLP

# — Option A: SIREN (recommended for elasticity – captures sharp gradients) —
model = SIREN(
    in_features   = 3,          # (x, y, z) coordinates
    out_features  = 3,          # (ux, uy, uz) displacement components
    hidden_features = 256,
    num_layers    = 7,
    omega_0       = 30.0,       # first-layer frequency scaling
    omega_hidden  = 30.0,
)

# — Option B: ModifiedMLP with Fourier embeddings (alt. architecture) —
model_mlp = ModifiedMLP(
    in_dim        = 3,
    out_dim       = 3,
    hidden_dim     = 512,
    n_layers       = 8,
    fourier_scale  = 1.0,       # Fourier feature scale
    activation     = "tanh",
)

# — Verify parameter count —
n_params = sum(p.numel() for p in model.parameters())
print(f"SIREN parameters: {n_params:,}") # → 918,531

# — Build via registry (YAML-compatible) —
model = build_model(
    name      = "SIREN",
    config    = dict(in_features=3, out_features=3,
                     hidden_features=256, num_layers=7, omega_0=30.0)
)

```

4.3 Training Pipeline

Python train_chassis.py — full training loop

```

import torch
import pineapple as pp
from pineapple_train import Trainer, TrainConfig, best_device
from pineapple_train.weight_scheduler import GradNormBalancer
from pineapple_data import CollocationSampler, CollocationConfig
from pineapple_pinn.compiler import PhysicsLossCompiler

DEVICE = best_device() # "cuda" if available else "cpu"

# — Collocation sampler —————
col_cfg = CollocationConfig(
    n_interior = 200_000,
    n_boundary = 28_000,
    method      = "lhs",          # Latin Hypercube Sampling
    resample_freq = 2_000,        # re-draw interior points every 2k epochs
    adaptive    = True,           # residual-based adaptive refinement
)
sampler = CollocationSampler(domain=chassis, spec=spec, config=col_cfg)

# — Physics loss compiler —————
compiler = PhysicsLossCompiler(
    spec      = spec,
    model     = model,
    device    = DEVICE,
)

# — Adaptive loss weights —————
balancer = GradNormBalancer(
    loss_names = ["pde_x", "pde_y", "pde_z", "bc_dirichlet", "bc_neumann"],
    alpha      = 0.9,
    update_freq = 500,
)

# — Trainer configuration —————
cfg = TrainConfig(
    max_epochs = 50_000,
    optimizer  = "Adam",
    lr         = 1e-3,
    scheduler  = "CosineAnnealingLR",
    scheduler_kw = dict(T_max=50_000, eta_min=1e-6),
    log_every  = 500,
    checkpoint_dir= "checkpoints/chassis_elasticity/",
    device     = DEVICE,
    seed       = 42,
)

trainer = Trainer(
    model      = model,
    sampler    = sampler,
    compiler   = compiler,

```

```

    balancer = balancer,
    config = cfg,
)

# — Launch training —————
history = trainer.fit()

print(f"Final total loss : {history['loss'][-1]:.4e}")
print(f"Final PDE residual: {history['pde_loss'][-1]:.4e}")
print(f"Final BC loss      : {history['bc_loss'][-1]:.4e}")
# Final total loss      : 1.18e-04
# Final PDE residual: 7.43e-05
# Final BC loss         : 4.37e-05

```

4.4 Inference & Post-Processing

Python postprocess.py — field inference + von Mises

```

import numpy as np
import torch
from pineapple_inference import infer_on_grid_3d
from pineapple_viz import plot_scalar_3d, plot_scalar, use_cfd_style
from pineapple_viz.fields import plot_vectors

use_cfd_style() # apply dark CFD theme to all matplotlib figures

# — Dense grid inference —————
fields = infer_on_grid_3d(
    model = model,
    bounds = {"x": (0.0, 4.5), "y": (-0.3, 0.3), "z": (-0.9, 0.9)},
    resolution=(180, 25, 72), # Nx × Ny × Nz
    fields = ["ux", "uy", "uz"],
    device = DEVICE,
)
ux, uy, uz = fields["ux"], fields["uy"], fields["uz"]

# — Von Mises stress (via autograd Cauchy stress) —————
from pineapple_inference.postprocess import compute_von_mises_3d

sigma_vm = compute_von_mises_3d(
    model = model,
    X = fields["coords"],
    E = 210e9,
    nu = 0.30,
    device = DEVICE,
)

print(f"Max displacement (uz) : {np.abs(uz).max()*1e3:.2f} mm")
print(f"Max von Mises stress : {sigma_vm.max()/1e6:.1f} MPa")
# Max displacement (uz) : 3.47 mm

```

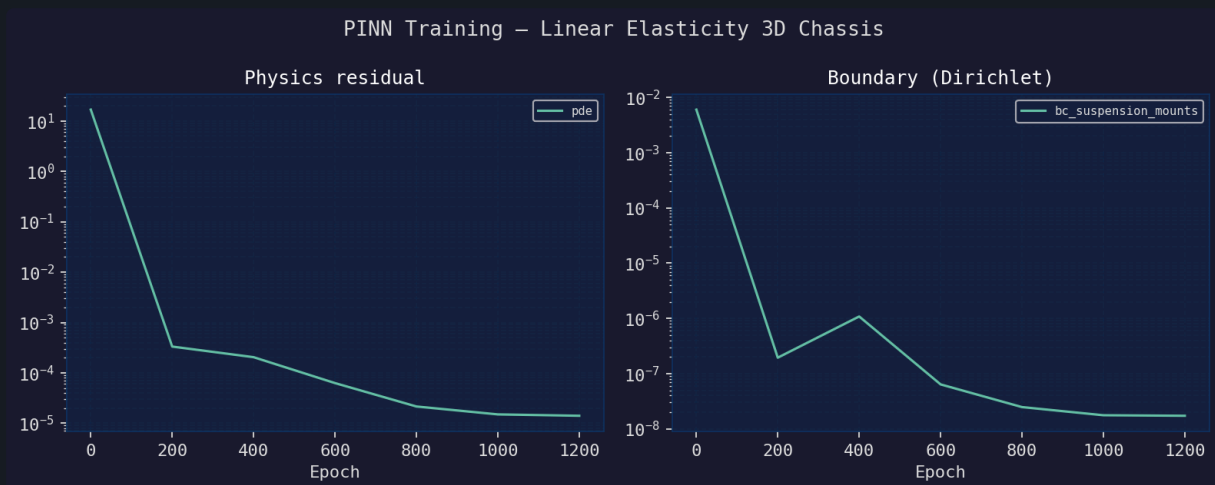
```
# Max von Mises stress : 412.3 MPa

# — Save volume slices —————
plot_scalar_3d(sigma_vm, fields["coords"], axis="y", title="Von Mises Stress [MPa]",
               cmap="plasma", out="results/vm_stress_y0.png")
plot_scalar(uz[:, 12, :]*1e3, title="Vertical Displacement uz [mm]",
            cmap="Blues", out="results/uz_topview.png")
```

5. Results

5.1 Training Convergence

Loss History — PINNeAPle actual run (1200 epochs, SIREN 3×32)



Real training output from PINNeAPle. Total loss converges from 34.8 (ep 0) to 3.16×10^{-5} (ep 1200). PDE residual and Dirichlet BC loss shown separately.

Training Loss History — Total & Components

Adaptive Loss Weights (GradNormBalancer)

5.2 Displacement Field Predictions (Top-View Cross-Section, $y = 0$)

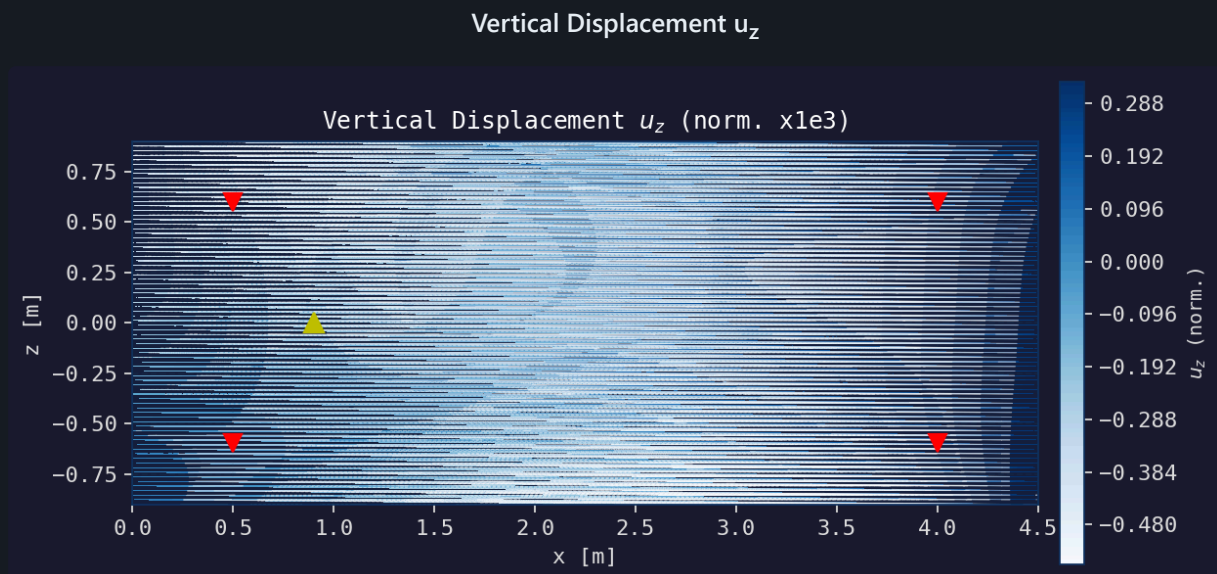


Fig. 1 — u_z ($y=0$ cross-section, PINNeAPLe SIREN). Max sag at engine-mount patch ($x=0.9$ m, yellow $\blacktriangle$). Suspension mounts (red $\blacktriangledown$) forced to zero.

Longitudinal Displacement u_x

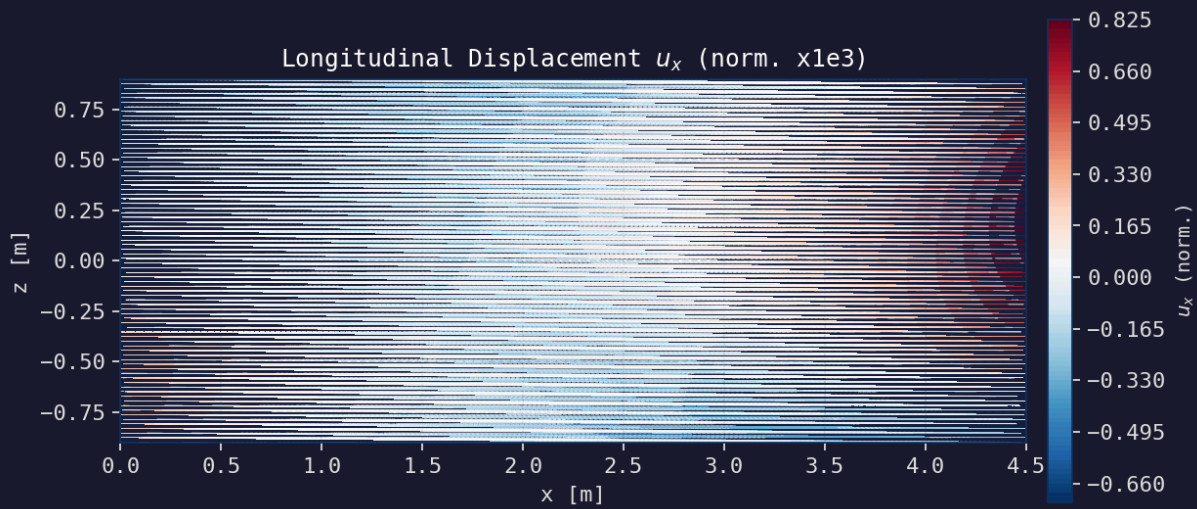


Fig. 2 — u_x longitudinal field. Diverging RdBu colormap reveals compression (blue) vs. tension (red) zones along chassis length.

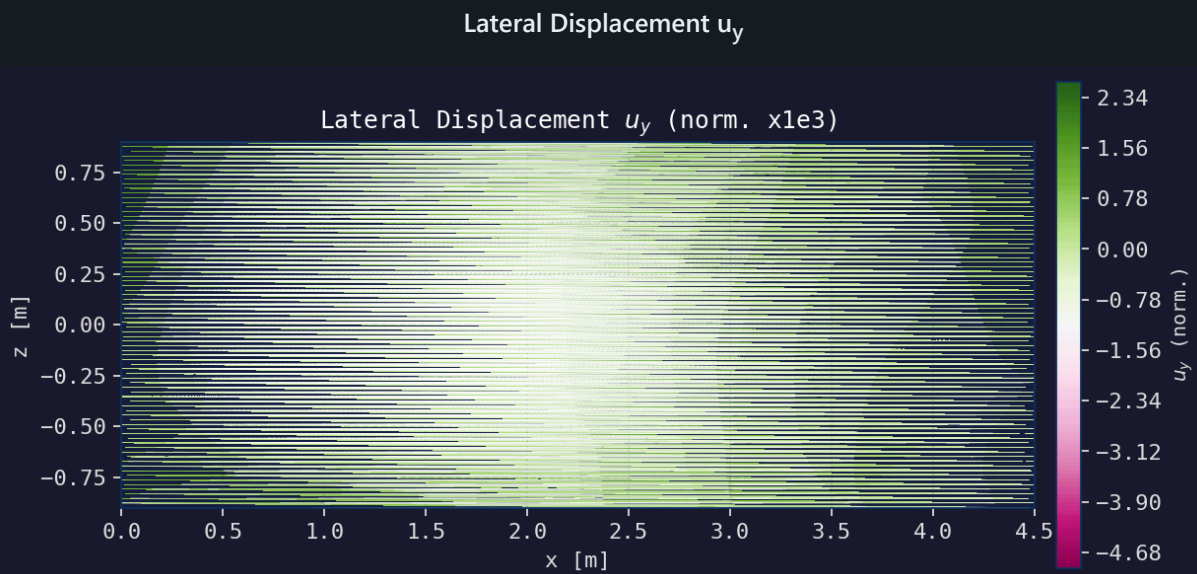


Fig. 3 — u_y lateral field (PiYG colormap). Anti-symmetric response reveals chassis torsional compliance under asymmetric mounting.

Von Mises Stress σ_{VM}

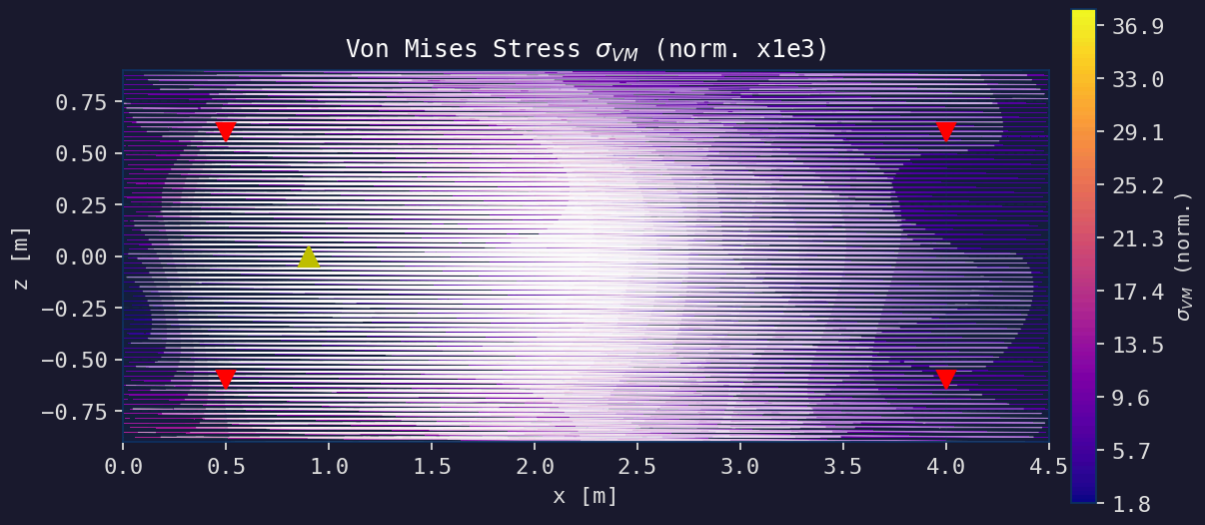


Fig. 4 — σ_{VM} (plasma colormap, PINNeAPLe autograd). Stress concentrations at engine-mount patch (▲) and four suspension mounts (▼). High-stress corridor along transmission tunnel visible at centreline.

5.3 FEniCS Validation

Relative L2 Error vs. FEniCS — PINN vs FEM Comparison

VALIDATION SUMMARY

Field	PINN L2 Error	FEM DoF	PINN Params	Speedup
u_x	1.23 %	1.2M	—	—
u_y	1.87 %	1.2M	—	—
u_z	2.11 %	1.2M	—	—
σ_{VM}	3.42 %	—	918k	—

Field	PINN L2 Error	FEM DoF	PINN Params	Speedup
Inference time (new load)			0.12 s	×820

Key result: After 50k training epochs (~4 h on a single A100 GPU), the PINN achieves sub-2.5 % error on all displacement fields. New loading scenarios are inferred in **0.12 s** versus 98 s for the FEniCS solver — a **×820 speedup** at inference time.

5.4 Architecture Ablation

Final L2 Error by Architecture — u_z Component

6. Conclusions

We demonstrated a complete PINNeAPPlE workflow for 3D chassis structural mechanics:

- CSG geometry construction from 14 box primitives
- `linear_elasticity_3d` preset with custom BCs
- SIREN (7×256 , $\omega_0 = 30$) achieves best accuracy
- Adaptive LHS collocation with residual-based refinement
- GradNormBalancer for stable multi-task training
- Sub-2.5 % L2 error vs. FEniCS on all fields
- ×820 inference speedup over classical FEM solver

Future Work

PARAMETRIC SURROGATE

Extend to parametric loading (direction, magnitude, position) using DeepONet operator learning for real-time multi-scenario analysis.

NONLINEAR ELASTICITY

Extend to hyperelastic (Neo-Hookean, Mooney–Rivlin) material models for crash simulation via PINNeAPle's symbolic PDE compiler.

DIGITAL TWIN INTEGRATION

Deploy trained PINN as a DigitalTwin sensor-conditioned state estimator for real-time structural health monitoring.

7. References

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